

TI-99 Console Tester Public Beta 0.7

This is the current public beta for the 16 KiB W27C512 modification of Geoff Trott's TI-99/4 and TI-99/4A Console Tester.

Which file to program

Program `HEXDIAG07.BIN` into the W27C512 used by the tested 16 KiB modification. It is the complete 65,536-byte image with the extension at `>6000->7FFF`, the fixed core at `>E000->FFFF`, and unused space filled with `>FF`.

Use `HEX8K07.BIN` for the first electrical/compatibility check. It repeats the fixed core in all eight physical W27C512 banks so A13, A14, and A15 cannot select different contents.

`HEXCORE7.BIN` and `HEXEXT07.BIN` are the two raw 8 KiB component images for developers. They are not complete W27C512 programmer files.

Before installation

1. Read `MANUAL07.PDF` and `HARDWARE.PDF`.
2. Compare the selected file with `SHA256.TXT`.
3. Program the entire device and run the programmer's verify pass.
4. Check the modification with the board unpowered.
5. Begin with the repeated-core image if the decoder or wiring is unproven.

The VRAM and memory tests are destructive. Do not run the diagnostic with unsaved machine state.

Qualification

The `HEXDIAG07.BIN` bytes passed the static acceptance checks and the internal 59-case emulator regression suite. Beta 0.7 was exercised on a stock TI-99/4A, a TI-99/4QI V2.2, a GROMless TI-99/4A through the LOAD recovery path, and a TI-99/4 fitted with a Pico9918.

License and credit

Source and build tooling are Apache-2.0. Original documentation and illustrated modification material are CC BY 4.0. The attributed MegaDemo-derived material has the separate permission boundary described in the repository's `docs/PROVENANCE.md`.

Credit **Jon Guidry (hexbus)** and link to the original project: github.com/hexbus/ti99-console-tester.